Causal Prototype-Inspired Contrast Adaptation for Unsupervised Domain Adaptive Semantic Segmentation of High-Resolution Remote Sensing Imagery

Jingru Zhu, Ya Guo, Geng Sun, Liang Hong, *Member, IEEE*, and Jie Chen, *Member, IEEE*

***Abstract*— Semantic segmentation of high-resolution remote sensing imagery (HRSI) suffers from the domain shift, resulting in poor performance of the model in another unseen domain. Unsupervised domain adaptive (UDA) semantic segmentation aims to adapt the semantic segmentation model trained on the labeled source domain to an unlabeled target domain. However, the existing UDA semantic segmentation models tend to align pixels or features based on statistical information related to labels in source and target domain data and make predictions accordingly, which leads to uncertainty and fragility of prediction results. In this article, we propose a causal prototype-inspired contrast adaptation (CPCA) method to explore the invariant causal mechanisms between different HRSIs domains and their semantic labels. It first disentangles causal features and bias features from the source and target domain images through a causal feature disentanglement (CFD) module. Then, a causal prototypical contrast (CPC) module is used to learn domain invariant causal features. To further de-correlate causal and bias features, a causal intervention (CI) module is introduced to intervene on the bias features to generate counterfactual unbiased samples. By forcing the causal features to meet the principles of separability, invariance, and intervention, CPCA can simulate the causal factors of source and target domains, and make decisions on the target domain based on the causal features, which can observe improved generalization ability. Extensive experiments under three cross-domain tasks indicate that CPCA is remarkably superior to the state-of-the-art methods.**

***Index Terms*— Causal view, contrastive learning, counterfactuals, disentangled representation, high-resolution remote sensing imagery (HRSI), semantic segmentation, unsupervised domain adaptation.**

I. Introduction

DRIVEN by advances in aerospace and sensor technology, an increasing number of high-resolution remote sensing imagery (HRSI) can be captured globally, which have rich spatial details and potential semantic contents. Identifying the semantic category of each pixel in HRSI has become an important task of remote sensing intelligent interpretation in the community. At present, the semantic segmentation of HRSI has been widely applied to urban planning [1], [2], [3], intelligent transportation [4], [5], disaster prediction [6], [7], [8], and agricultural production [9], [10].

In the past decade, deep learning methods, especially deep convolutional neural networks (DCNNs), have shown excellent performance in semantic segmentation tasks of HRSI. Compared to traditional machine learning methods for semantic segmentation, such as support vector machines [11], random forests [12], and conditional random fields [13], DCNN-based methods like FCN [14], U-Net [15], SegNet [16], Deeplab series [17], [18], [19], can capture multilevel and multiscale semantic information and finer grained context information, demonstrating great potential in feature representation and pattern recognition. However, due to the limitation of fixed receptive fields in convolutional operation, DCNN-based methods lack the ability to model global contextual information and long-range dependencies. Even though self-attention-based methods can alleviate the above problems [20], [21], they obtain global information based on the aggregation of local features obtained by DCNN, making it difficult to obtain accurate global context information in complex HRSI scenes. Recently, inspired by the success of transformer [22] in extracting global contextual relationships, transformer-based methods have gradually been applied to semantic segmentation of HRSI [23], [24], [25], and have shown excellent performance in extracting long-range dependencies of geo-objects and global context information.

Most DCNN-based semantic segmentation models assume that the data distribution of the training and testing sets is the same. However, different remote sensing image domains often face the problem of a "data distribution gap" mainly due to the discrepancy in sensor type, imaging conditions, and shooting location. Facing this situation, traditional semantic segmentation models suffer from significant performance drops and fail to achieve out-of-distribution (OOD) generalization [26], [27]. To address this problem, unsupervised domain adaptive (UDA) methods aim to transfer the knowledge learned from the labeled source domain to the unlabeled target domain.

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Currently, some UDA works relying on feature alignment have achieved remarkable performance for semantic segmentation by minimizing adversarial learning losses [27], [28], [29], [30] or contrastive learning loss [31] between source and target domains. Specifically, adversarial learning-based methods typically involve two-players: a feature extractor and a discriminator. The former uses a CNN-based network to extract features from images of source and target domains, while the latter determines whether the input features are from the source or target domain. The model is well-trained when two-players reach a Nash equilibrium in a min–max game through adversarial learning, where the feature extractor can produce aligned and indistinguishable feature representations between two domains. In order to ensure the consistency of the local joint distribution when the alignment of global distribution, some adversarial learning-based works attempt to take the category-wise information into account during the adversarial learning. For example, a category-level adversarial network [29] was proposed to weight the adversarial loss for different features by emphasizing the role of category-level feature alignment in reducing domain shifts. Chen et al. [32] adaptively reduced the attention of classifier on category-level aligned features while increasing the attention on category-level unaligned features by considering category-certainty maps. FADA [33] proposed a fine-grained adversarial learning strategy to incorporate class information into the discriminator for aligning features at a fine-grained level. Unlike adversarial learning, a contrastive adaptation framework [31] was proposed to simultaneously consider the alignment of intraclass features and the utilization of interclass relationships. Inspired by the success of the vision transformer (ViT) [34] in image visual recognition, the transformer has now been applied to the UDA semantic segmentation of images [35].

These methods tend to align image styles or features based on learned statistical information. However, the statistical dependence-based methods are prone to show uncertainty and vulnerability in the prediction results due to the interference of chaotic phenomena [36]. Therefore, in order to better realize high-performance semantic segmentation between different domains of HRSI, merely considering the statistical dependency between variables is insufficient, but an underlying causal model [37], [38], [39], [40], [41]. For example, the tree in HRSI with RGB band presents green, showing high statistical dependence between the "green" and "tree" label in the semantic segmentation task of HRSI, which could easily mislead the model to make a wrong prediction when the tree presents red in HRSI with IRRG band in the target domain. After all, the essential characteristics of a tree are the rounded crown and the regular textural features in HRSI, rather than color.

In order to explore the invariant causal mechanism between different HRSI domains and semantic labels, in this article, we address the cross-domain semantic segmentation task of HRSIs from a causal view, and formalize it as a structure causal model (SCM) [42] by examining on the causalities among the following six factors: unobserved causal factors C, unobserved noncausal factors U_S and U_T, observed images X_S and X_T, and ground truth labels Y. Specifically, we assume that the semantically relevant information in images X_S and X_T serve as causal factors, which are usually domain-invariant information, e.g., information about the content structure of images corresponding to the semantic labels. Whereas category-independent information as noncausal factors, which are usually domain-specific information, e.g., style information in an image. Fig. 1(a) illustrates the SCM of UDA semantic segmentation of HRSI, where $U_S \rightarrow X_S \leftarrow C$ ($U_T \rightarrow X_T \leftarrow C$) represents that each observed X_S (X_T) is composed of domain-invariant C and domain-specific U_S (U_T). $U_S \dashleftarrow\dashrightarrow C$ ($U_T \dashleftarrow\dashrightarrow C$) denotes a pseudo-correlation between the U_S (U_T) and C, which is usually caused by unobserved confounders. $X_S \rightarrow F_S \rightarrow Y$ and $X_T \rightarrow F_T \rightarrow Y$ represent that existing UDA methods usually learn the images embedding F_S (F_T) based on the observed X_S (X_T), and make the prediction Y based on the F_S (F_T). $C \rightarrow Y$ represents that C is the only parent variable that determines the generation of the semantic label Y. However, existing UDA methods typically ignore the causal mechanism between source and target domains.

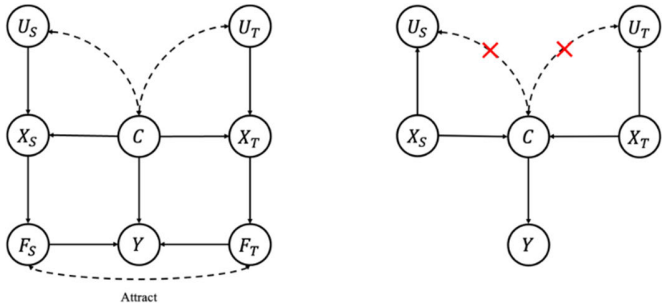


Fig. 1. SCM of UDA semantic segmentation of HRSI. The solid arrow indicates that the parent node causes the child one; while the dashed arrow means there exists statistical dependence. (a) SCM of domain adaptation. (b) SCM of the proposed method.

According to d-connection theory [43]: two variables are interdependent if they are connected by at least one unblocked path. We can find two paths that will lead to a pseudo-correlation between U_S (U_T) and Y in Fig. 1(a): 1) $U_S \rightarrow X_S \rightarrow F_S \rightarrow Y$ or $U_T \rightarrow X_T \rightarrow F_T \rightarrow Y$ and 2) $U_S \dashleftarrow\dashrightarrow C \rightarrow Y$ or $U_T \dashleftarrow\dashrightarrow C \rightarrow Y$. In order to make prediction Y solely dependent on C, we need to intercept the above two unblocked paths. To this end, in order to intercept path 1), we hope to separate the potential domain-invariant C and the domain-specific U_S (U_T) from X_S (X_T) and predict Y solely based on C, namely, $U_S \leftarrow X_S \rightarrow C \rightarrow Y$ or $U_T \leftarrow X_T \rightarrow C \rightarrow Y$. In order to intercept path 2), one possible solution is to make C and U_S (U_T) irrelevant as we cannot change the connection between C and Y, namely, $U_S \dashleftarrow\dashrightarrow \times \dashleftarrow\dashrightarrow C$ or $U_T \dashleftarrow\dashrightarrow \times \dashleftarrow\dashrightarrow C$. Fig. 1(b) illustrates the SCM of our proposed method. Unfortunately, since causal/noncausal factors are generally unobservable and unformulated, we cannot disentangle causal and noncausal factors directly from the input images [44], [45]. Therefore, in this article, we introduce the disentangled representations learning to characterize the causal features, and make them

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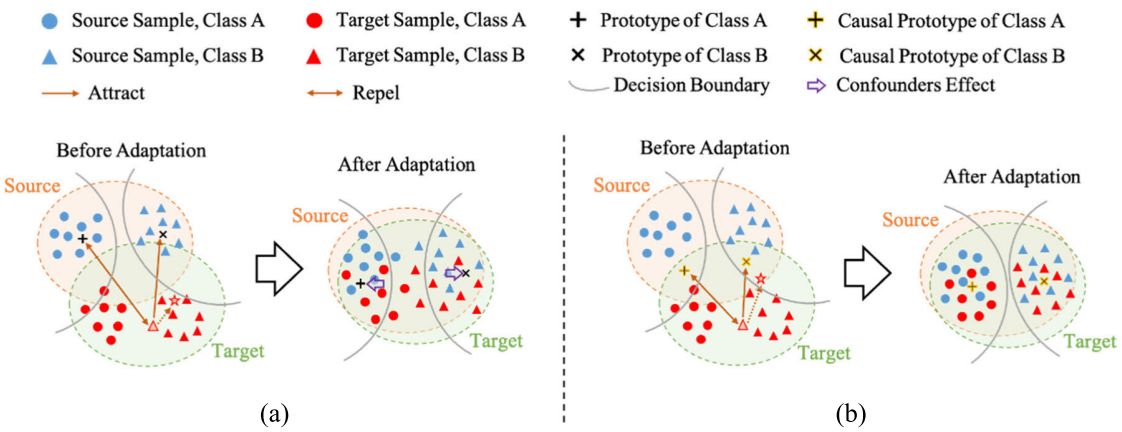


Fig. 2. Illustration of causal prototype modeling. (a) Without causal prototype modeling. (b) With causal prototype modeling.

satisfy the following three principles for replacing causal factors: 1) the causal factors should be separated from the noncausal factors; 2) the causal factors of source and target domains should be domain invariant; and 3) intervention on noncausal factors will not affect the causal factors to make decisions on labels.

Based on the above analysis, we proposed a causal prototype-inspired contrast adaptation (CPCA) method for the UDA semantic segmentation task of HRSIs. First, we design a causal feature disentanglement (CFD) module to disentangle the causal features and bias features from the original HRSIs to satisfy the first principle of causal factors. Notable, the causal features obtained by the CFD module are independent of the source and target domains, respectively, and are not domain-invariant. To further learn the domain-invariant causal features, we develop a causal prototypical contrast (CPC) module to bridge the discrepancy between source and target domain causal features, which can satisfy the second principle of causal factors. Finally, causal features should be predictively adequate for semantic labels, and its decision-making will not be affected when intervening on the bias features. Therefore, we introduce a causal intervention (CI) module to intervene on bias features to generate counterfactual unbiased samples for further decorating the causal and bias features, which can satisfy the third principle of causal factors. As shown in Fig. 2, compared with the traditional UDA model, our proposed model with causal prototype modeling can learn to characterize causal features during domain adaptation, which leads to a more compact domain-invariant feature space where features of the target domain appear correctly within the decision boundary.

The contributions of our work are as follows.

1) We introduce a causal view to formalize the cross-domain semantic segmentation problem for HRSIs and propose a CPCA algorithm to separate causal features from the original input HRSIs by disentangled representation learning (DRL) and force the causal features to replace causal factors by satisfying three principles that causal factors should possess in UDA task.
2) We propose a causal prototypical contrastive learning to bridge the discrepancy between the causal features of the source and target domain, motivating the model to obtain domain-invariant causal features.
3) A CI module is proposed to exert intervention on bias features to generate counterfactual unbiased samples, which further de-correlates causal features and bias features so that causal features are sufficiently predictive for semantic labels.

The remainder of this study is organized as follows. Section II summarizes the related work to our study. Section III introduces the proposed method in detail. Section IV discusses the experiment settings, comparative experimental result analysis, ablation experiments, and visualization of causal features and bias features. Section V summarizes our work.

II. Related Work

A. UDA for Semantic Segmentation

Existing methods for UDA semantic segmentation tasks can be divided into three groups, including image-to-image translation, adversarial learning, and self-training. The image-to-image translation methods aim to eliminate the style differences between source and target domains by using a generative adversarial network (GAN) framework to perform style transfer between source and target domains [46], [47]. Adversarial learning methods focus on generating invariant spaces shared by the source and target domains through adversarial loss of the feature space [27] or the output space [30]. Some studies have reinforced the learning of invariant spaces by introducing category information [29], [32], or multiscale information [28] in adversarial learning. Furthermore, Jiang et al. [31] used contrastive learning instead of adversarial learning to align the intraclass distribution while distancing the interclass distribution between the source and target domains. On the other hand, inspired by semi-supervised learning (SSL), self-training adaptative methods focus on assigning high-confidence pseudo-labels to the target domain and involving them in the training of semantic segmentation models. An iterative self-training method [48] was proposed to iteratively generate pseudo-labels on target domains and retrain the semantic segmentation model with these labels. ProDA [49] introduced a prototypical pseudo-label denoising method to correct the soft pseudo-labels of the target

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domain, enabling the network to learn from high-confidence denoised pseudo-labels of the target domain during training. A transformer-based model named DAFormer [35] first applied Transformers to UDA and revealed its potential for semantic segmentation of UDA. Specifically, it consists of a Transformers encoder and a context-aware fusion decoder and employs three crucial training strategies to avoid overfitting to the source domain.

In the field of remote sensing, the performance of semantic segmentation models is usually affected by the differences in sensor types, imaging conditions, and imaging regions. To solve this problem, domain adaptation methods have been gradually applied to the semantic segmentation task of HRSIs. To consider the consistency of local distribution along with the alignment of global distribution, category-certainty attention is proposed to focus on category-level unaligned regions in remote sensing images [32]. [50] introduced an intermediate domain to connect source and target domains, and utilized it to achieve knowledge transfer between the source domain and the target domain. MemoryAdaptNet [51] introduced an invariant feature memory module to learn domain-invariant features and aggregate them into the current image, further enhancing the representation of the current features by employing a category attention-driven aggregation module. A multiprototype learning method [52] was proposed to model complex intraclass and interclass relationships for effectively alleviating the effect caused by intradomain and interdomain differences in UDA semantic segmentation of HRSIs.

However, these current UDA semantic segmentation models are committed to learning the statistical information related to semantic labels from source and target domain images, and aligning or transforming distributions based on it to make predictions for target domain images. Obviously, these models are susceptible to uncertainties and show vulnerability and noninterpretability in predicting labels when facing complex and variable remote sensing image scenes. In contrast, we adopt a UDA semantic segmentation method of HRSI from a causal view with a structure casual model, which focuses on exploring the inherent causal mechanism of cross-domain remote sensing images by learning domain-invariant causal features to provide better generalization ability and interpretability.

B. Causal Learning

Statistical learning models can learn the correlation of data. It can perform better in independent and identically distributed (i.i.d.) settings when enough i.i.d. data are observed. However, these methods typically perform poorly when the data does not meet the i.i.d condition or when the modeling scenery is complex. In contrast, causal learning [43] strives to discover more essential causal structures from the data beyond correlations. According to Judea Pearl's causal ladder [53], causal reasoning can be divided into three levels: association, intervention, and counterfactual, which can describe the variant and invariant parts of the data distribution by modeling the human processes of observation, action, decision-making, and reflection. Therefore, over the past few years, causal learning has received increasing attention in the areas of data generation, domain generalization, and model interpretability [54], [55]. For example, a graph surgery estimator relying on causal directed acyclic graph (DAG) was proposed for recognizing and representing the invariance required for changes due to dataset shifts [56]. Heinze-Deml and Meinshausen [57] viewed the data-generating process from a causal perspective and categorized the latent data-generating factors into invariant and style features. For the domain generalization task, MatchDG [57] proposed a structural causal model to formalize the domain generalization problem and illustrated the importance of intraclass variation modeling for generalization. CIRL [45] exploited dimension-wise representations to mimic causal factors based on theoretical properties and then reconstructed invariant causal mechanisms to address domain generalization. To explain black-box visual classifiers, Panda et al. [58] formulated a causal extension to the paradigm of instance-wise feature selection and obtained a subset of input features that has the greatest causal effect on the model's output. Liu et al. [59] conducted a comprehensive review of existing causal reasoning methods for visual representation learning and pointed out the importance of causal reasoning in visual representation learning.

These studies are usually based on causality to obtain invariant causal mechanisms or to recover causal features in order to improve the OOD generalization problem or the interpretability problem of visual representation learning. However, few existing methods are designed for UDA semantic segmentation of HRSIs. Since the observable variables and latent causal variables in the structural causal model are different between the UDA task and DG task, it is necessary to design a causal structure model for the task of UDA semantic segmentation of HRSIs and to mine the causal invariant mechanism of HRSIs in cross-domain scenarios.

C. Disentangled Representation Learning

DRL aims to identify and reveal hidden representations in observational data with human-like generalization capabilities, attempting to break the black-box properties of existing end-to-end deep learning models. Existing studies [60], [61] have demonstrated the potential of DRL in modeling human learning and understanding of the world, mainly because DRL encourages the representations to carry interpretable semantic information with independent factors and shows significant potential for representing invariance [62], [63], integrity [64], [65], and generalization [66], [67]. In view of this, CRL is now widely applied in computer vision [68], [69], [70], [71], [72], natural language processing [73], [74], [75], [76], [77], and graphical learning [78], [79], [80]. The most typical DRL approaches are based on generative models such as variational auto-encoder (VAE) [81] or GAN [82]. For example, IDVAE [83] designed a new identifiable double VAE framework for learning optimal representation of latent space to learn disentangled representations. A disentangled VAE [67] was proposed to disentangle category-distilling and category-dispersing factors from visual and semantic features, respectively, to avoid nonrecognition-related information in zero-shot learning. Aiming to address age-invariant face verification tasks, DR-RGAN achieves disentangled

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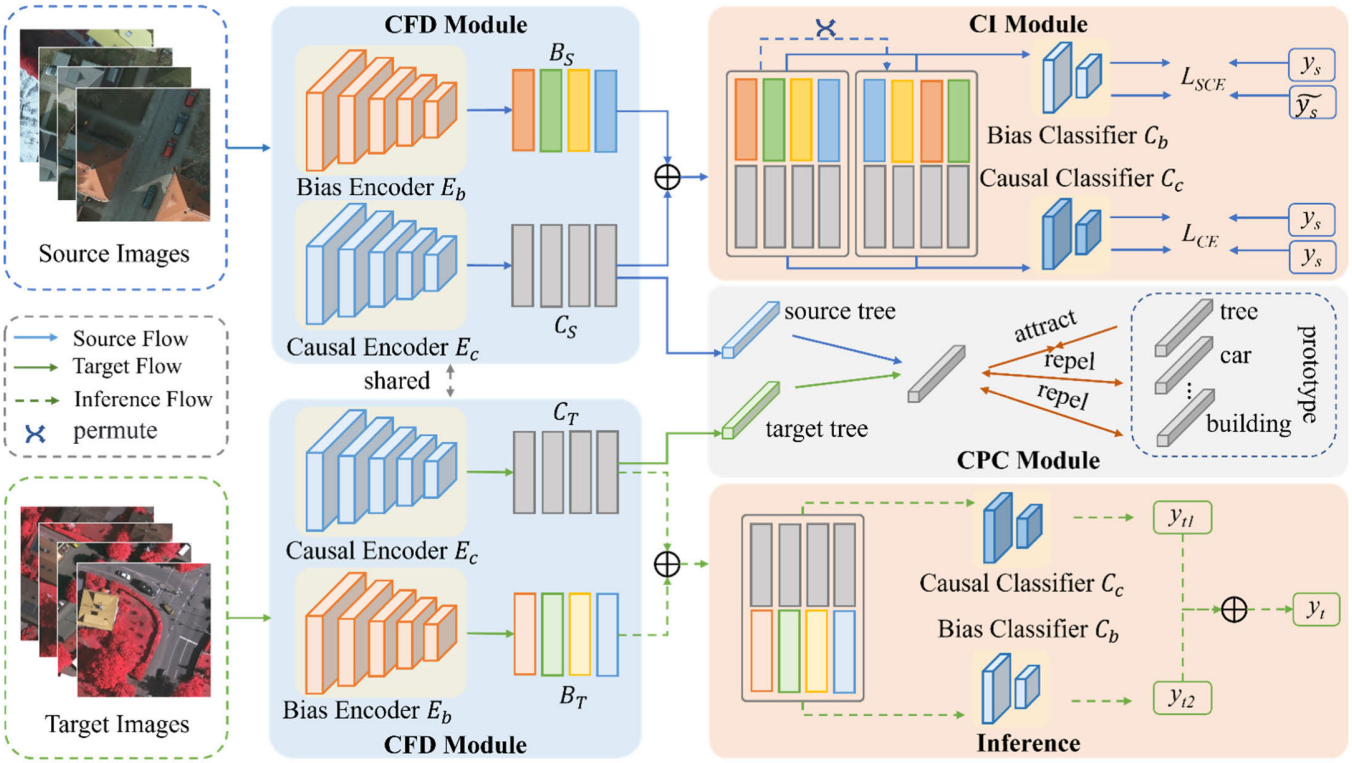


Fig. 3. Structure of CPCA.

representation by representing the facial features without age interference. These approaches show greater potential for learning interpretable representations of images. In addition, causal inference-based DRL is also widely used. For example, a VAE-based causal DRL framework named CausalVAE [64] was proposed to recover the latent factors with semantics and structure via a causal DAG. Reddy and Balasubramanian [84] presented a causal perspective for learning trustworthy and transferable disentangled representations and captured the generation process of latent variable representation learning models. These approaches aim to encourage networks to learn potential causal disentangled representations from a causal perspective. Inspired by this, we propose a DRL method for better solving the domain shift problem of UDA semantic segmentation of HRSI by disentangling domain-invariant causal features from source- and target-domain images.

III. Methodology

In this section, we focus on the causal prototype-inspired UDA semantic segmentation methods for HRSIs. Given the source domain images X_S with K-class pixel-level labels Y_S and the target domain images X_T without pixel-level labels, our goal is to construct domain-invariant causal mechanisms from labeled source domain images and unlabeled target domain images based on the above three principles of causal factors, so that the model can be well generalized in the target domain. As shown in Fig. 3, our proposed CPCA consists of three modules: the CFD module, the CPC module, and the CI module. First, X_S and Y_S are input into the CFD module to obtain causal features C_S, bias features B_S of the source domain, and causal features C_T, bias feature B_T of the target domain, respectively. Second, for obtaining domain-invariant causal features between source and target domains, a CPC module is proposed to bridge the discrepancy between C_S and C_T by contrastive learning between C_S (C_T) and causal prototypes. Finally, after the domain-invariant causal features are well-learned, we utilize the CI module to swap interventions on B_S and B_T in each min-batch to get counterfactual unbiased samples, which can enforce the classifier to learn the unbiased information and promote the predictive adequacy of causal features for semantic labels.

A. CFD Module

In order to eliminate the spurious correlation between bias factors B_S (B_T) and label Y, and intercept the paths 1) and 2) in SCM of Fig. 1(a), we need to disentangle potential causal and noncausal factors from the observed images X_S and X_T. However, since both causal and noncausal factors are hidden variables rather than directly observed variables, it is difficult to directly disentangle the images X_S and X_T into $X_S = f(C, U_S)$ and $X_T = f(C, U_T)$, where $f(\cdot)$ is the disentanglement function. Inspired by recent works on DRL [84], [85], [86], in this article, we disentangle the causal and bias features from the observed images from the perspective of DRL and attempt to substitute causal factors with causal features that force them to satisfy the three principles of causal factors.

In this section, we introduce the CFD module to disentangle causal and bias features for carrying out the separation of causal features from bias features. Specifically, as shown in Fig. 3, the CFD module consists of a causal encoder E_c, and

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a bias encoder E_b, which are used to extract causal and bias features from input images, respectively. Given a mini-batch X_S and X_T, we input X_S and X_T into E_c to produce source causal features C_S and target causal features C_T. Meanwhile, the X_S and X_T are inputted into E_b simultaneously, and mapped into bias features B_S and B_T. E_c and E_b share the same network structure and can be any DCNN-based network. In this article, we use ResNet-50 [87] pretrained on source datasets as the backbone of E_c and E_b. Finally, through the CFD module, we disentangle the X_S into C_S and B_S, and disentangle the X_T into C_T and B_T. Intuitively, E_c and E_b can focus on the different parts of the structure information of the original image.

B. CPC Module

The C_S and C_T obtained by the CFD module are independent of the source and target domains, respectively. To further achieve better generalization in the target domain, the causal features should be domain-invariant. Therefore, in order to learn domain-invariant causal features so that the causal features satisfy the second principle of causal factors, in this section, we use the CPC module to construct category-level domain-invariant causal prototypes p and bridge the discrepancy between the C_S (C_T) and the p through contrastive learning. Specifically, the CPC module consists of three steps: causal prototypes initialization, causal prototypes contrastive learning, and causal prototypes updating.

1) Causal Prototypes Initialization: Before domain adaptation, we first train the semantic segmentation model on the labeled source domain dataset in a fully supervised manner and use the cross-entropy loss function to constraint the training of the segmentation model as follows:

$$L_{\text{seg}}(X_S, Y_S) = -E_{(x_s, y_s)\sim(X_S, Y_S)} \sum_{i=1}^{K} y_s^{(i)} \log \hat{y}_s^{(i)} \tag{1}$$

where $L_{\text{seg}}(X_S, Y_S)$ is the segmentation loss, $x_s \in X_S$ represents the images from the source domain, and $y_s \in Y_S$ represents the ground truth corresponding to x_s, $\hat{y}_s$ represents the predicted label corresponding to x_s, and K represents the number of classes.

After obtaining the model trained on the labeled source domain, we initialize the p with deep features of the source domain

$$p = \left\{ p_1, p_2, \ldots, p_K \,\middle|\, p_k = \frac{\sum_{n=1}^{N_s} C_n^s \mathbb{I}\left[Y_n^s = k\right]}{\sum_{n=1}^{N_s} \mathbb{I}\left[Y_n^s = k\right]} \right\} \tag{2}$$

where N_S is the number of source domain images, $C_n^s \in R^d$ is the source causal feature vector with dimension d, k is the index of categories number K, $I[Y_n^s = k]$ is an indicator function, which equals to 1 if $Y_n^s = k$ and 0 otherwise.

2) Causal Prototypes Contrastive Learning: In order to bridge the discrepancy between C_S and C_T and obtain domain-invariant causal features, we use the contrastive learning paradigm that considers both interclass and intraclass relationships instead of the adversarial learning paradigm [27], [28], [30] for feature alignment in this article. First, we compute the similarity S_S^k (S_T^k) between C_S (C_T) and causal prototype p_k of each category k, respectively,

$$S_S^k = \frac{\exp(p_k \cdot C_S/\tau)}{\sum_{k=1}^{K} \exp(p_k \cdot C_S/\tau)} \tag{3}$$

$$S_T^k = \frac{\exp(p_k \cdot C_T/\tau)}{\sum_{k=1}^{K} \exp(p_k \cdot C_T/\tau)} \tag{4}$$

where τ represents the temperature that controls the sharpness of the generated probability distribution. Then, we calculate the cross-entropy loss between the S_S^k (S_T^k) and semantic labels

$$L_S = -\sum_{k=1}^{K} y_S^k \log S_S^k \tag{5}$$

$$L_T = -\sum_{k=1}^{K} \tilde{y}_T^k \log S_T^k \tag{6}$$

where y_S^k is the ground truth label of the source domain and $\tilde{y}_T^k$ is the predicted label of the target domain. The final causal prototypes contrastive loss is

$$L_{\text{CC}} = L_S + L_T. \tag{7}$$

CPC module can narrow the discrepancy of causal features in the same category, while widening the discrepancy of causal features in different categories between source and target domains, thus achieving the learning of domain-invariant causal representation.

3) Causal Prototypes Updating: To enhance the domain-invariant representation of causal prototypes, we integrate the causal information of source and target domains into p with a moving average scheme. Specifically, we first update the causal features C_k for each category k based on $S = \{S_S^k, S_T^k\}s$ as

$$C_k' = \sum_{i=1}^{L} \frac{1 - S^i}{\sum_{j=1}^{L} (1 - S^j)} \cdot C_k^i \tag{8}$$

where L represents the number of pixels labeled as k in C, $C_k^i = \{C\mathbb{I}[Y = k]\} \in C_k$.

Finally, we calculate the updated prototype of each category k by leveraging the moving average

$$p_k' = (1 - m) \cdot p_k + m \cdot C_k' \tag{9}$$

where m is the momentum, and we employ a polynomial annealing policy to update it

$$m_t = \left(1 - \frac{t}{T}\right)^{\alpha} \cdot \left(m_0 - \frac{m_0}{100}\right) + \frac{m_0}{100}, t \in [0, T] \tag{10}$$

where m_t represents the value of m at the tth iteration, T represents the total number of training iterations, and α and m_0 are default parameters and set to 0.9.

C. CI Module

According to the above description, the causal and bias features from source and target domain images can be disentangled using the CFD module, and the causal features of source and target domains can conform to the domain-invariant principle using the CPC module. However, the causal features C_S (C_T) and the bias feature B_S (B_T) are disentangled from

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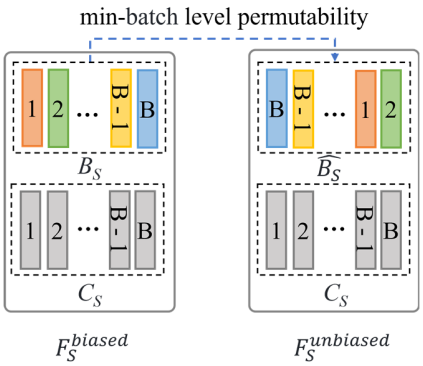


Fig. 4. Counterfactual instances generation.

the biased source domain (target domain) images, and there is a statistical correlation between C_S (C_T) and B_S (B_T). Given Judea Pearl's Ladder of causality theory [43]: the causal factors should satisfy the requirements of intervention and counterfactuals, we argue that the causal features C_S (C_T) should remain invariant to interventions on B_S (B_T), when causal features and bias features are not confounded, i.e., adjustment formula: P [C_S | do(B_S)] or P [C_T | do(B_T)], where do($\cdot$) denotes intervention on variables.

Therefore, to further decorrelate C_S (C_T) with B_S (B_T), in this article, we intervene on the B_S by swapping in each min-batch to generate counterfactual features as shown in Fig. 4. More specifically, for the bias features B_S of each min-batch $\{1, 2, \ldots, B-1, B\}$, where B represents the number of min-batch sizes, we perform random permute at the min-batch to generate the permuted bias features $\hat{B}_S$. Then, we concatenate it with the causal features C_S in the same min-batch to obtain the counterfactual features $F_S^{\text{unbiased}} = [C_S; \widehat{B_S}]$.

Since C_S and $\widehat{B_S}$ in F_S^{unbiased} may be a random combination of different images, they have less correlation than C_S and B_S in $F_S^{\text{biased}} = [C_S; B_S]$ where both are from the same images. Based on the principle that the causal factors should remain predictive adequacy for interventions on noncausal factors: the classification of the F_S^{unbiased} should make the same prediction as the preintervention features F_S^{biased}. Therefore, for the F_S^{biased} and F_S^{unbiased}, we use the following loss function to train the causal classifier C_c:

$$L_C = L_{C_1} + L_{C_2} \tag{11}$$

$$L_{C_1} = \text{CE}\left(C_c\left(F_S^{\text{biased}}\right), y_s\right) \tag{12}$$

$$L_{C_2} = \text{CE}\left(C_c\left(F_S^{\text{unbiased}}\right), y_s\right) \tag{13}$$

where CE represents the cross-entropy loss and y_s represents the source label.

Meanwhile, to ensure that bias encoder E_B focus on the bias information, and decorrelation C_S and B_S, we further swap label y_s as $\hat{y}_s$ along with $\hat{B}_S$ to train the bias classifier C_b

$$L_B = L_{B_1} + L_{B_2} \tag{14}$$

$$L_{B_1} = \text{SCE}\left(C_b\left(F_S^{\text{biased}}\right), y_s\right) \tag{15}$$

$$L_{B_2} = \text{SCE}\left(C_b\left(F_S^{\text{unbiased}}\right), \hat{y}_s\right) \tag{16}$$

where SCE (.) represents the symmetric cross entropy [88].

Together with the causal prototypes contrast loss, the total loss function is defined as

$$L = L_{\text{CC}} + L_C + L_B. \tag{17}$$

It is worth noting that in order to ensure the quality of the counterfactual features $\widehat{B_S}$ and $\widehat{B_T}$, in the early stage of training, we only use L_{CC}, L_{C_1}, and $L_{B_1}$to train the model. After τ iterations, we use L to train the model.

D. Self-Training

After the causal-prototype contrast adaptation stage, we can further improve the performance of UDA semantic segmentation of HRSI by self-training strategy following previous works [89]. Given the target images $X_T = \{x_n^t\}_{n=1}^{N^t}$, where N^t is the number of target images. We input each image x_t^n to the causal-prototype contrast adaptation model, and to obtain the predicted label set $\tilde{Y}_T = \{\tilde{Y}_n^t \in R^{H\times W}\}_{n=1}^{N^t}$ and a ranked confidence set $\delta_k = \left[\delta_k^1, \delta_k^2, \cdots, \delta_k^l\right]$ of category k, where l represents the length of confidence set belonging to category k, it can be calculated as follows:

$$l = \sum_{n=1}^{N_t}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbb{I}[\tilde{Y}_n^t = k] \tag{18}$$

where $\tilde{Y}_n^t$ is the predicted label for the image x_n^t, $\mathbb{I}[\tilde{Y}_n^t = k]$ is an indicator function, which equals to 1 if $\tilde{Y}_n^t = k$ and 0 otherwise. Then, we use the threshold σ_k to determine whether a pixel is associated with category k

$$\tilde{P}_n^t = \left\{\tilde{Y}_n^t \in R^{H\times W} \,\middle|\, \delta_k \geq \sigma_k\right\} \tag{19}$$

where $\tilde{P}_n^t$ represents the pseudo label for the image x_n^t, $\sigma_k = l * \eta$, η is the percentage, and we set it as 0.9.

IV. Experiments and Result Analysis

A. Experimental Settings

1) Datasets: The data domain gaps between HRSIs are mainly reflected in the differences in color texture, scale size, and shape characterization of geo-object classes due to differences in sensor type, resolution size, imaging location, or imaging weather. To verify that our method can better handle these common distribution differences of HRSIs in semantic segmentation tasks, we conduct experiments on two HRSI datasets provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) WG II/4: Potsdam dataset [90] and Vaihingen dataset [91]. Specifically, both datasets provide a certain number of high-resolution true orthophoto tiles (TOPs) with their corresponding pixel-by-pixel semantic labels, which consist of six general land cover categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter (clutter/background). In addition, there are data distribution differences in terms of sensor type, resolution size, and imaging location between these two datasets. For example, due to the sensor type is different, the Potsdam dataset provides four bands: near-infrared (IR), red (R), green (G), and blue (B), while Vaihingen dataset only provides three bands: IR, R, and G, so we can select three bands (such

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TABLE I
DOMAIN GAPS BETWEEN THREE UDA TASKS

Domain gaps	Datasets			Tasks		
	POT(IR-R-G)	POT (R-G-B)	VAI (IR-R-G)	POT(IR-R-G) → VAI (IR-R-G)	POT (R-G-B) → VAI (IR-R-G)	VAI (IR-R-G) → POT (IR-R-G)
Spatial resolution (m)	0.05	0.05	0.09	√	√	√
Geographic location	Potsdam	Potsdam	Vaihingen	√	√	√
Scene	city	city	small village	√	√	√
Imaging sensors	IR-R-G	R-G-B	IR-R-G		√	

as R, G, and B bands) from the Potsdam dataset that are different from the Vaihingen dataset and combine them to simulate the domain differences under different sensor types. In terms of resolution size, the Potsdam dataset contains 38 TOPs with a size of 6000 × 6000 and a resolution of 5 cm, and the Vaihingen dataset contains 33 TOPs with a size of approximately 2494 × 2064 and a resolution of 9 cm. In terms of imaging locations, the Potsdam dataset describes an urban scene with a crowded residential pattern, while the Vaihingen dataset describes a small rural scene with a sparse layout. Therefore, setting up a cross-domain semantic segmentation task between these two datasets can fully test the ability of our method to handle the common domain differences in remote sensing images.

2) Task Settings: In this article, we set up three UDA semantic segmentation tasks: POT (IR-R-G) → VAI (IR-R-G), VAI (IR-R-G) → POT (IR-R-G), and POT (R-G-B) → VAI (IR-R-G), where POT (IR-R-G) → VAI (IR-R-G) represents the Potsdam dataset with a band combination of IR-R-G as the source domain and the Vaihingen dataset with a band combination of IR-R-G as the target domain; VAI (IR-R-G) → POT (IR-R-G) represents the Vaihingen dataset with a band combination of IR-R-G as the source domain and the Potsdam dataset with a band combination of IR-R-G as the target domain; POT (R-G-B) → VAI (IR-R-G) represents the Potsdam dataset with a band combination of R-G-B as the source domain and the Vaihingen dataset with a band combination of IR-R-G as the target domain. The domain discrepancy information on these three UDA semantic segmentation tasks is shown in Table I.

In the model training phase, we crop the Potsdam dataset and the Vaihingen dataset into image patches according to the size of 512 × 512 and the step size of 256 where the training set of the Potsdam dataset contains 11 616 patches, and the test set contains 1694 patches; the training set of Vaihingen dataset contains 1296 patches, and the test set contains 112 patches. Other detailed settings are the same as in [51]. In this case, the source domain images, semantic labels, and target domain images are involved in the training process, while the semantic labels of the target domain are not accessible during the training process.

3) Implementation Details: We implement our network using the PyTorch toolbox on a single RTX 3090 GPU with 24 GB memory. All experiments were implemented in the PyCharm Development Environment. The key parameters for network training are shown in Table II.

4) Evaluation Metric: We use three common evaluation metrics: F_1 score, over accuracy (OA), and mean intersection and union (IoU) to evaluate the performance of different UDA semantic segmentation methods.

TABLE II
KEY TRAINING PARAMETERS INVOLVED IN THIS WORK

Parameters	Source-only model	CPCA model	Self-training model
Optimizer	SGD	SGD	SGD
Initial learning rate	5 × 10−4	1 × 10−3	5 × 10−4
Learning rate momentum	0.9	0.9	0.9
Learning rate weight decay	5 × 10−4	5 × 10−4	5 × 10−4
Batch size	8	2	8
Training times of iteration	20000	50000	20000

Specifically, the F_1 score is formulated as follows:

$$F_1 = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{20}$$

where the Precision and Recall can be formulated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{22}$$

Moreover, OA is as shown as follows:

$$\text{OA} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{23}$$

For mIoU, we first compute the IoU for each category separately and then average them to obtain mIoU. The following equation shows how to compute the IoU for each category:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{24}$$

where TP denotes the number of true positive pixels, FP denotes the number of false positive pixels, and FN denotes the number of false negative pixels, respectively.

B. Comparative Experimental Result Analysis

To verify the validity of the proposed method, we perform comparison experiments on a source-only model without UDA and nine state-of-the-art UDA methods without considering the causality (MCD_DA, AdaptSegNet, AdvEnt, FADA, SDCA, MUCSS, Zhang's, ProCA, and DAFormer), in which MUCSS, Zhang's are the state-of-the-art UDA methods proposed in the remote sensing semantic segmentation, and we directly used the quantitative evaluation results reported in the original paper.

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TABLE III
QUANTITATIVE EVALUATION RESULTS (%) OF DIFFERENT UDA MODELS ON THE POT (IR-R-G)→ VAI (IR-R-G) TASK

Methods		F_1 score						OA	MA	mIoU
		Imp. surf.	Building	Low veg.	Tree	Car	Clutter			
Source-only		72.85	74.40	56.42	67.96	46.76	28.49	67.19	60.55	43.58
State-of-the-art methods	MCD_DA [92]	77.00	80.30	54.94	70.48	54.28	nan	70.20	57.66	46.63
	AdaptSegNet [28]	77.69	85.99	60.97	74.06	52.01	26.43	73.70	64.00	48.65
	AdvEnt [30]	77.55	84.42	52.93	74.23	53.98	30.51	74.43	65.29	51.37
	FADA [93]	78.77	87.84	60.84	78.31	57.82	37.76	76.60	67.72	52.55
	SDCA [94]	81.80	88.55	56.47	77.29	64.19	31.20	76.71	67.77	52.79
	MUCSS [95]	66.13	76.77	55.97	73.14	51.09	45.65	61.43	-	45.38
	Zhang's [96]	80.13	86.65	**64.16**	71.90	61.94	31.34	66.02	-	52.03
	ProCA [31]	82.42	90.31	63.68	78.49	56.54	37.45	77.48	69.25	54.37
	DAFormer [35]	80.59	90.40	59.37	79.53	67.27	39.72	77.85	74.82	55.70
(Ours CPCA)		**84.92**	**91.76**	61.99	**79.75**	**68.10**	**60.18**	**80.18**	**76.94**	**60.75**

MCD_DA [92]: A UDA method that attempts to align source and target domain distributions by considering task-specific decision boundaries.

AdaptSegNet [28]: A UDA method employing multilevel adversarial learning in the output space for pixel-level semantic segmentation.

AdvEnt [30]: A UDA method for semantic segmentation via adversarial training approach based on the entropy of semantic predictions.

FADA [93]: A UDA method for class-level feature alignment by fine-grained adversarial learning strategy.

SDCA [94]: A UDA method that aligns the pixel-wise representation guided by semantic distribution.

MUCSS [95]: A UDA method under multiple weakly-supervised constraints to minimize the data shift of remote sensing images.

Zhang's [96]: A UDA method exploiting curriculum-style and local-to-global adaptation to address the cross-domain problem of remote sensing images.

ProCA [31]: A UDA method that aligns the class-centered distribution by prototypical contrastive learning.

DAFormer [35]: A UDA method based on the transformers encoder and a context-aware fusion decoder, which reveals the potential of transformers for cross-domain semantic segmentation.

With the exception of DAFormer, which uses the MiT-B5 [97] encoder, all comparative methods use the DeepLab-v3 with ResNet-101 pretrained on ImageNet as the semantic segmentation network. Limited by the Memory of GPU, our model uses the DeepLab-v3 with ResNet-50. All methods carry out experiments on these three tasks: POT (IR-R-G) → VAI (IR-R-G), VAI (IR-R-G) → POT (IR-R-G), and POT (R-G-B) → VAI (IR-R-G).

1) Comparative Studies on POT (IR-R-G) → VAI (IR-R-G): The quantitative evaluation results of the compared domain adaptation methods on the POT (IR-R-G) → VAI (IR-R-G) task are presented in Table III. From the results, due to the domain shifts in spatial resolution, geographic location, and scene, the source-only model has the worst performance with OA, MA, and mIoU values of 67.19%, 60.55%, and 43.58%, respectively. The nine state-of-the-art UDA comparison methods achieve varying degrees of performance improvement compared to the source-only model due to accessing images in the target domain and using a domain-adaptive strategy. However, these methods tend to align pixels or features based on the statistical information learned from images and their corresponding labels, which leads to the uncertainty and noninterpretability of the prediction results when facing complex and variable remote sensing image scenes, limiting the performance of UDA semantic segmentation for HRSIs. Therefore, the best-performing model DAformer based on transformer among the comparison methods only achieves the performance with OA, MA, and mIoU values of 77.85%, 74.82%, and 55.70%, respectively. Considering the invariant causal mechanism between different remote sensing image domains and semantic labels in the process of domain adaptation, our method CPCA achieves the best performance with OA, MA, and mIoU values of 80.18%, 76.94%, and 60.75%, respectively. Compared with the source-only model, the CPCA shows a performance improvement of 12.99% on OA, 16.39% on MA, and 17.17% on mIoU. Compared with the comparison method DAformer with the best performance, the CPCA presents an improvement of 2.33%, 2.12%, and 5.05% on OA, MA, and mIoU, respectively.

In Table III, the F_1 scores of CPCA in impervious surfaces, buildings, trees, cars, and clutter realize the best accuracies of 84.92%, 91.76%, 79.75%, 68.10%, and 60.18%, respectively, and achieve the improvement of 12.07%, 17.36%, 11.79%, 21.34%, and 31.69% compared to the Source-only model. Notable, the CPCA achieved the highest improvement of 60.18% compared to the Source-only model and 20.46% compared to the best-performing model DAformer on the clutter which contains multiple geo-objects such as water bodies, containers, tennis courts, and swimming pools. This result demonstrates that the CPCA has higher performance and better generalization for such categories with complex features, multiple semantics, and diverse scenes, which benefits from the fact that the CPCA mitigates the interference of complex appearance features and irrelevant external factors on the performance of semantic segmentation by capturing the invariant causal mechanism between images and semantic categories in different remote sensing image domains.

We selected six typical remote sensing scenarios in the test set of the Vaihingen dataset for comparing the cross-domain semantic segmentation performance on the POT (IR-R-G) → VAI (IR-R-G) task. Fig. 5 is the qualitative results of six

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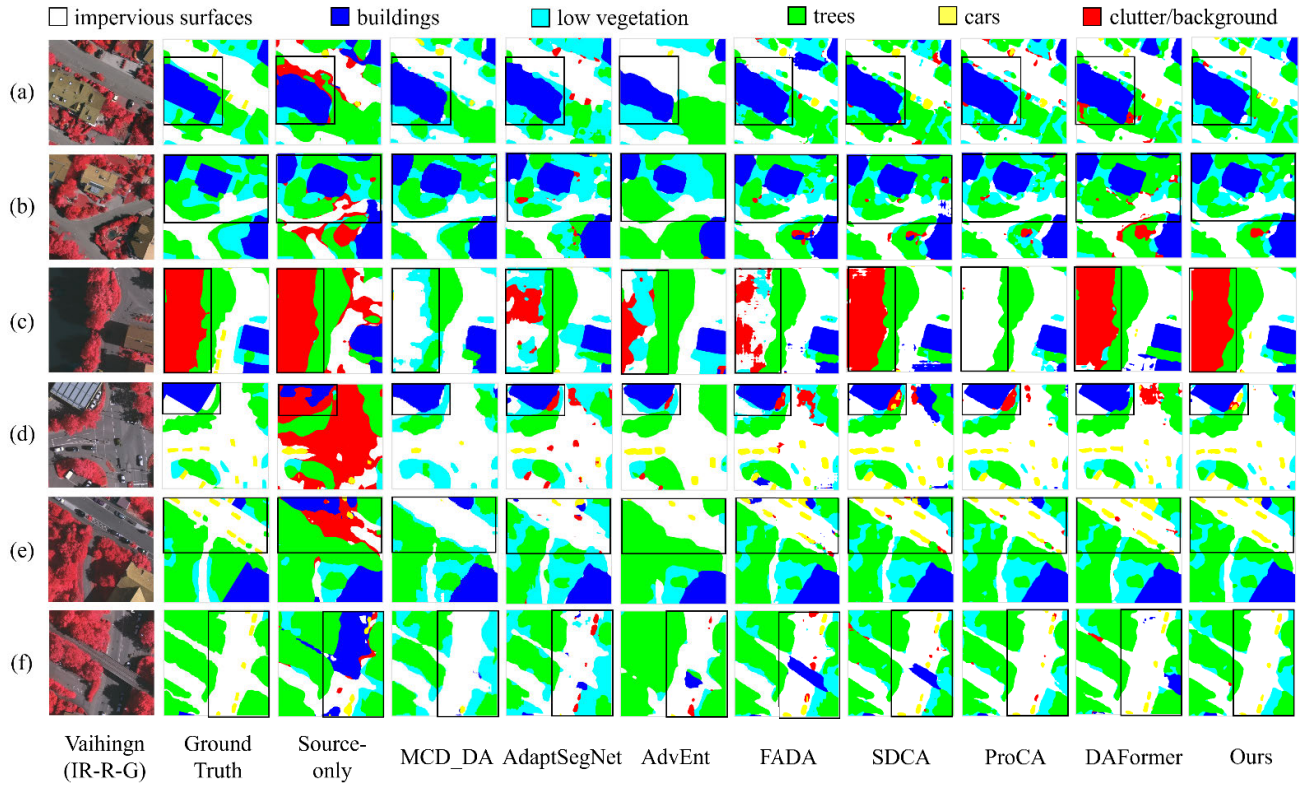


Fig. 5. Qualitative results on the POT (IR-R-G) → VAI (IR-R-G) task.

typical remote sensing scenarios (a)–(f) in the Vaihingen dataset, where the semantic segmentation performance of the source-only model is poor due to the domain discrepancy between source and target domains, e.g., there is serious confusion between clutter and other categories, between low vegetation and trees, and between impervious surfaces and buildings. After domain adaptation, the confusion problem has been improved, but there is still confusion between clutter and impervious surfaces, and cars. By mining the causal relationship between two domain images and semantic labels, compared with other methods, our method achieves better semantic segmentation performance and improves the confusion problem between categories, such as between clutter and buildings, low vegetation, and trees in Fig. 5(a) and (b); between clutter and impervious surfaces in Fig. 5(c), between clutter and cars in Fig. 5(d)–(f). In summary, compared to other UDA semantic segmentation methods without considering the causality, our method achieves the best semantic segmentation performance in the POT (IR-R-G) → VAI (IR-R-G) task.

2) Comparative Studies on POT (R-G-B) → VAI (IR-R-G): The quantitative evaluation results of the compared domain adaptation methods on the POT (R-G-B) → VAI (IR-R-G) task are presented in Table IV. Except for domain shift in spatial resolution, geographic location, and scene layout, the discrepancy on the POT (R-G-B) → VAI (IR-R-G) task also includes imaging sensors. Due to this domain shift, the source-only model also has the worst performance with OA, MA, and mIoU values of 57.38%, 48.07%, and 33.58%, respectively. By considering the narrowness of domain shift between the source and target domains, nine comparative state-of-the-art UDA methods all achieve better segmentation performance than the source-only model, but there is still significant room for improvement. The proposed CPCA achieves the best performance with MA, and mIoU values of 71.75%, and 47.67%, respectively. Compared with the source-only model, our CPCA shows the performance improvement of 12.38% on OA, 23.68% on MA, and 14.09% on mIoU. Compared with the comparison method ProCA with the best performance, CPCA presents the improvement of 7.68%, and 2.04% on MA and mIoU, respectively. This demonstrates that feature alignment methods based on causal invariance are more effective in narrowing the distribution discrepancy between different domains.

Fig. 6 shows the qualitative results of the POT (IR-R-G) → VAI (IR-R-G) task with six typical remote sensing scenarios (a)–(f) in the Vaihingen dataset. From Fig. 6, the semantic segmentation performance of the source-only model does not yet meet the requirements of practical applications because of the existence of semantic blurring and classification confusion between geo-objects. After domain adaptation, the semantic segmentation performance of all comparative state-of-the-art UDA methods is improved compared with the source-only model, but there are still problems of classification confusion and boundary-blurring, for example, buildings and clutter are confused with each other in Fig. 6(a)–(c), and cars and clutter are confused in Fig. 6(d)–(f). However, the method proposed in this article can better alleviate the above problems and achieve the best semantic segmentation performance. This is because the

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TABLE IV
QUANTITATIVE EVALUATION RESULTS (%) OF DIFFERENT UDA MODELS ON THE POT (R-G-B)→ VAI (IR-R-G) TASK

Methods		F_1 score						OA	MA	mIoU
		Imp. surf.	Building	Low veg.	Tree	Car	Clutter			
Source-only		64.91	63.58	43.54	63.97	40.44	7.94	57.38	48.07	33.58
State-of-the-art methods	MCD_DA [92]	69.75	75.46	51.62	50.85	35.05	0.00	61.75	48.30	36.52
	AdaptSegNet [28]	68.31	78.23	46.27	63.73	40.44	8.09	63.13	51.08	37.55
	AdvEnt [30]	69.49	75.87	32.96	73.72	49.79	6.39	62.04	52.80	37.64
	FADA [93]	79.84	85.32	25.44	76.14	56.91	14.52	69.37	66.32	44.08
	SDCA [94]	70.16	87.05	41.42	74.89	47.79	30.25	69.48	59.99	44.38
	MUCSS [95]	61.33	83.00	42.17	70.66	57.88	13.88	54.82	-	39.93
	Zhang's [96]	76.89	84.81	**56.26**	68.10	57.15	18.64	60.31	-	46.13
	ProCA [31]	76.77	**88.71**	42.93	64.38	50.97	20.67	**71.39**	64.07	45.63
	DAFormer [35]	72.37	83.48	16.74	79.13	58.22	4.08	62.05	59.18	41.02
(Ours)		**81.06**	84.67	43.34	**81.50**	**60.08**	**39.64**	69.76	**71.75**	**47.67**

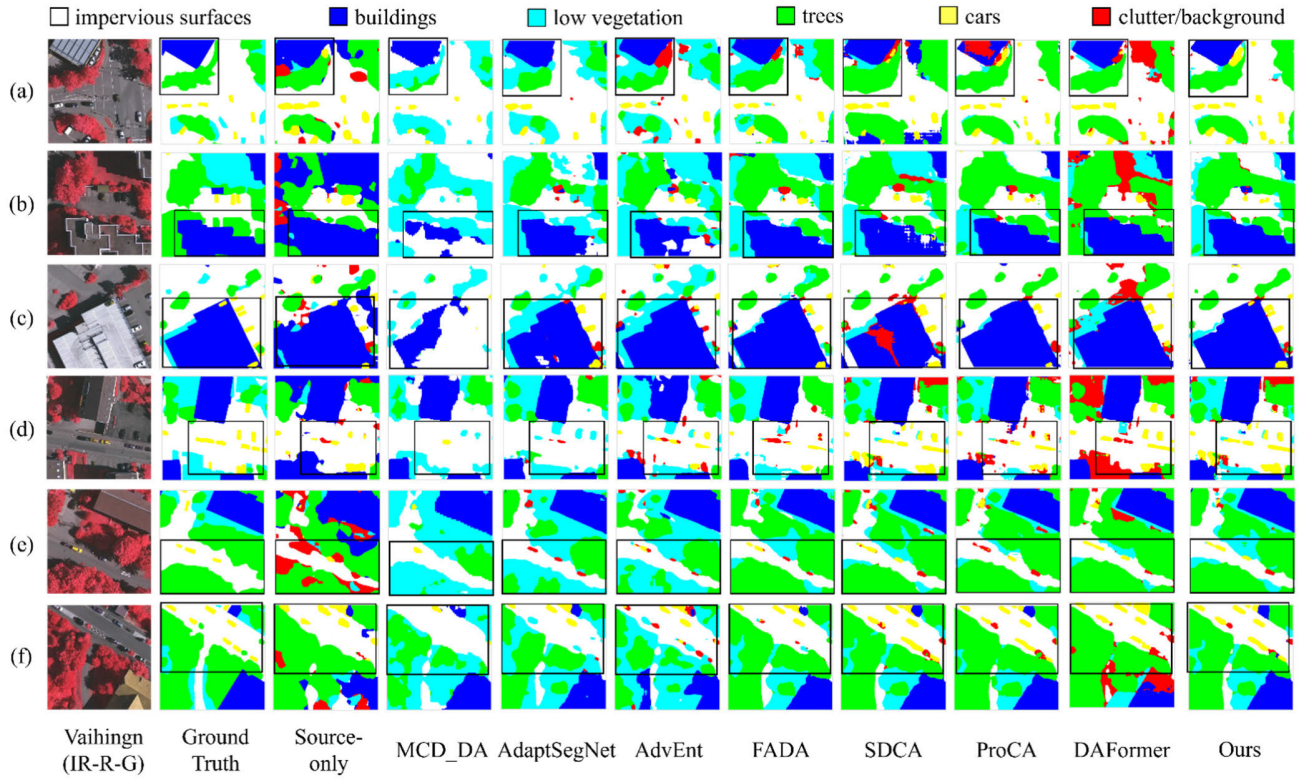


Fig. 6. Qualitative results on the POT (R-G-B) → VAI (IR-R-G) task.

category-level causal relationship between two domain images and semantic labels can be effectively mined through the contrastive learning of category-level causal features, which can better alleviate the problem of inaccurate semantic information and confusing classification.

3) Comparative Studies on VAI (IR-R-G) → POT (IR-R-G): The quantitative evaluation results of the compared domain adaptation methods on the VAI (IR-R-G) → POT (IR-R-G) task are presented in Table V. From Table V, the source-only model also has the worst performance with OA, MA, and mIoU values of 64.36%, 54.05%, and 41.22%, respectively. By considering the reduction of the domain shift between the source and target domains, nine comparative state-of-the-art UDA methods achieve better segmentation results than the source-only model, but there is still much room for improvement. The CPCA achieves the best performance with MA, and mIoU values of 63.01%, and 50.72%, respectively. Compared with the source-only model, CPCA shows a performance improvement of 8.82% on OA, 8.96% on MA, and 9.50% on mIoU. Notably, DAFormer achieved similar accuracy scores as our method CPCA, reflecting the advantages of the Transformer structure in feature extraction. Compared to the DAFormer, our method based on DeepLab-v3 with ResNet-50 still achieved 0.62%, and 0.08% improvement in MA and mIoU, respectively. This demonstrates that disentangling causal features from input images and then aligning them between the source and target domains from a causal perspective can effectively bridge the distribution shift between different domains, thus alleviating the impact of domain discrepancy on cross-domain semantic segmentation tasks.

From Table V, the F_1 scores of CPCA in impervious surfaces, buildings, cars, and clutter realize the best accuracies

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TABLE V
QUANTITATIVE EVALUATION RESULTS (%) OF DIFFERENT UDA MODELS ON THE VAI (IR-R-G)→ POT (IR-R-G) TASK

Methods		F_1 score						OA	MA	mIoU
		Imp. surf.	Building	Low veg.	Tree	Car	Clutter			
Source-only		71.32	69.45	60.97	**54.67**	69.10	5.47	64.36	54.05	41.22
State-of-the-art methods	MCD_DA [92]	75.67	76.72	63.53	39.97	60.68	0.00	67.04	56.42	43.71
	AdaptSegNet [28]	76.55	77.93	62.15	45.45	70.89	6.16	68.93	57.33	44.94
	AdvEnt [30]	76.65	82.64	63.73	32.94	71.74	2.76	69.59	58.03	45.43
	FADA [93]	79.04	82.89	69.31	29.02	82.75	5.82	71.20	59.63	46.75
	SDCA [94]	79.35	82.17	70.10	41.35	78.76	5.06	71.74	60.03	47.18
	MUCSS [95]	67.53	69.59	53.48	51.82	65.31	20.56	54.71	-	39.30
	Zhang's [96]	78.59	79.84	63.27	54.60	75.08	24.59	62.66	-	47.87
	ProCA [31]	77.81	81.20	**71.94**	37.90	81.78	15.56	71.90	60.52	48.20
	DAFormer [35]	79.46	**88.41**	70.45	35.49	82.69	16.63	**74.14**	62.39	50.64
(ours)		**81.51**	85.59	71.02	38.00	**84.45**	**26.68**	73.18	**63.01**	**50.72**

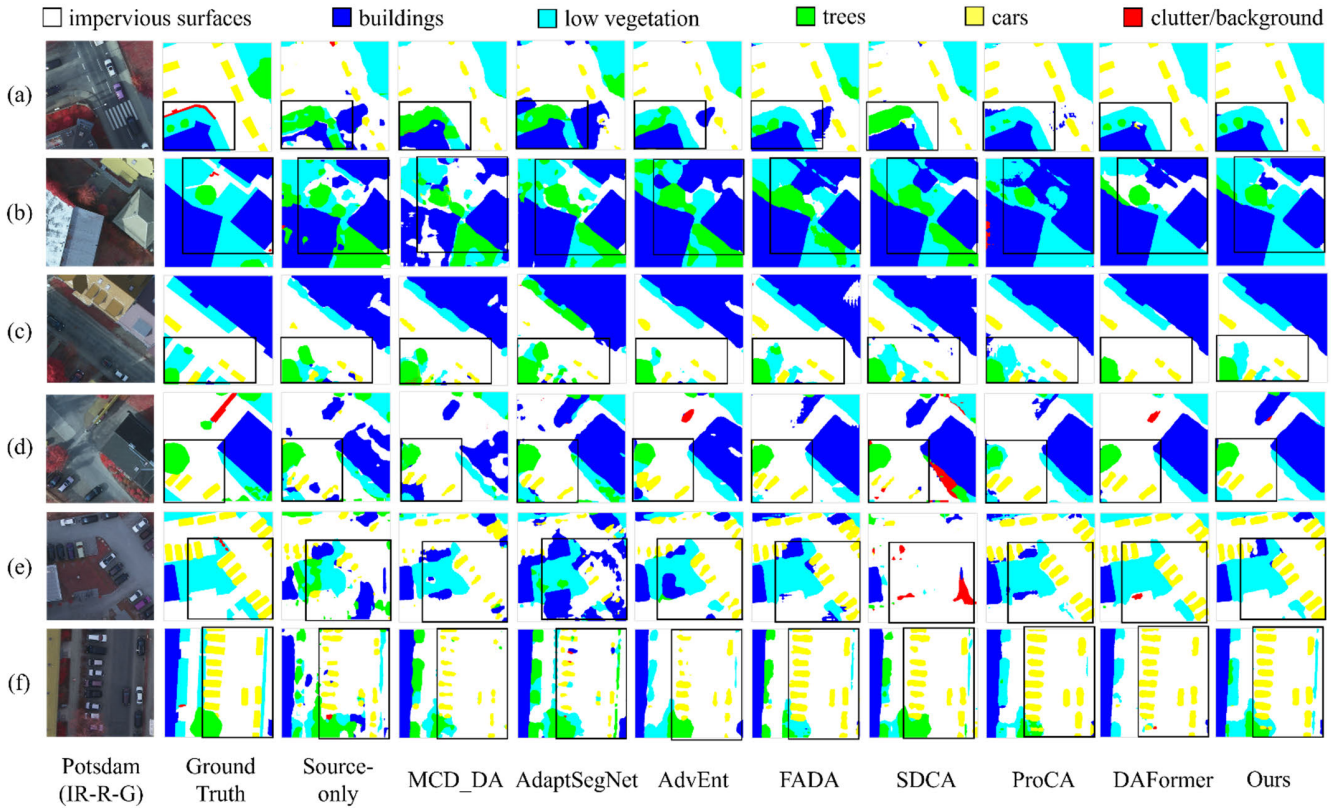


Fig. 7. Qualitative results on the VAI (IR-R-G) → POT (IR-R-G) task.

of 81.51%, 85.59%, 84.45%, and 26.68%, respectively, and achieve improvements of 10.19%, 16.14%, 15.35%, and 21.21% compared to the source-only model. It is worth noting that all domain adaptation methods, including the proposed one, show negative transfer to trees. This is mainly due to the fact that the trees in the Potsdam dataset are sparser and barer relative to the dense and tall trees in the Vaihingen dataset, resulting in objects under the trees not being shaded and presenting a more diverse visual appearance in remote sensing images. As a result, the models tend to predict the trees as other geo-object classes, such as cars, buildings, and impervious surfaces.

We selected six typical remote sensing scenarios in the Potsdam test set for comparing the cross-domain semantic segmentation performance on the VAI (IR-R-G) → POT (IR-R-G) task, Fig. 7 shows the qualitative results of six typical remote sensing scenarios (a)–(f) in the Potsdam dataset. From Fig. 7, the source-only model has poorer semantic segmentation results compared with ground truth due to the domain gap. The semantic segmentation results of the nine state-of-the-art UDA comparative methods are improved compared to the source-only model, but still suffer from confusion, boundary-blurring, and cavity of geo-object in the predicted mask, such as the confusion between impervious surfaces and buildings in Fig. 7(a), (b), and (e), as well as between low vegetation and trees in Fig. 7(a)–(c). While our method CPCA achieves the best semantic segmentation performance, which effectively alleviates the above problems, and achieves better segmentation performance on buildings, cars, and low vegetation. For example, CPCA effectively solved the cavitation problem of building in Fig. 7(b)–(d), and captured the long strip features of low vegetation more completely in Fig. 7(f). This shows

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TABLE VI
NUMBER OF PARAMETERS AND FLOPS OF ALL TESTED METHODS

Methods	MCD_DA	AdaptSegNet	AdvEnt	FADA	SDCA	ProCA	DAFormer	Ours
Parameters(M)	99.3	67.15	67.15	42.65	42.65	42.65	86.33	47.90
FLOPs(GMac)	228.23	568.36	284.18	178.81	178.81	178.81	116.5	204.02

TABLE VII
ABLATION STUDY RESULTS (%) ON THE POT (IR-R-G)→ VAI (IR-R-G) TASK

Source-only	CFD	CPC	CI	Self-training	OA	mIoU
√					67.19	43.58
	√				76.31	52.52
	√		√		76.86	53.82
		√			76.84	53.32
	√	√			77.37	54.42
	√	√	√		77.00	56.32
	√	√	√	ResNet-50	79.26	60.42
				ResNet-101	**80.18**	**60.75**

that our CPCA can better handle the VAI (IR-R-G) → POT (IR-R-G) task.

Table VI shows the statistics information of parameters and floating point operations (FLOPs) for all experimental methods. From Table VI, our method has relatively low parameters of 47.90M, which outperforms the methods of MCD_DA, AdaptSegNet, AdvEnt, and DAFormer. Although our method has moderate computational complexity due to the concatenation of causal and bias features, it still outperforms the methods of MCD_DA, AdaptSegNet, and AdvEnt.

C. Ablation Experiments

1) Effectiveness of Each Component: To evaluate the contribution of each module in the proposed CPCA, Table VII presents the ablation study results of the CFD, CPC, CI components, and self-training strategy on POT (IR-R-G) → VAI (IR-R-G) task. Compared to the source-only model, employing the CFD module improved the results by 9.0% in OA, and 8.94% in mIoU, respectively, which confirms the validity of the disentanglement of causal features. The further integration of the CI module improved OA and mIoU to 76.86% and 53.82%. Meanwhile, compared to the source-only model, employing the CPC improved the results by 9.65% in OA, and 9.74% in mIoU, respectively, which confirms the effectiveness of causal prototypical contrastive learning in aligning feature distribution. Notable, only employing the CPC module achieves better segmentation performance than only employing the CFD module with an improvement of 0.53% on OA, and 0.80% on mIoU, which indicates that pulling the feature distributions between the source and target domain is more effective than disentangling the causal feature from input images. Furthermore, simultaneously employing the CFD module and CPC module improved the OA and mIoU to 77.37% and 54.42%, which achieves further improvements of 1.06% on OA and 1.9% on mIoU, respectively, compared to the model only employing the CFD module, and achieves further improvements of 0.53% on OA, 1.1% on mIoU, respectively, compared to the model only employing the CPC module. The performance of OA and mIoU reaches 77.00% and 56.32%, respectively, when adding the CI module. Finally, we obtain the pseudo-label of the target domain based on the CPCA model and perform self-training on the target domain on the Deeplab-v3 model with ResNet-50 and ResNet-101 backbone, respectively. It achieved further improvements of 2.26% on OA and 4.1% on mIoU with ResNet-50 backbone, and 3.18% on OA and 4.43% on mIoU with ResNet-101 backbone.

2) CAM of Each Component: In order to visually show the importance of each component in the model, we visualize the class activation map (CAM) of features in different combinations of the model (such as CFD, CPC, CFD + CI, CFD + CPC, and CFD + CPC + CI) on the POT (IR-R-G) → VAI (IR-R-G) task. From Fig. 8, the model with the CFD component has deviations in the feature responses of each category. For example, the response to buildings involves impervious surfaces and cars, the response to cars is mistakenly located in buildings and trees, the response to impervious surfaces mistakenly involves buildings, the response to low vegetation involves impervious surfaces, and the response to trees is more responsive to low vegetation, resulting in poor semantic segmentation performance. The model with the CPC component responds more accurately than the model with the CFD component in the buildings, cars, and impervious surfaces categories, but still has some response bias in the categories of impervious surfaces and trees. Furthermore, the models with the CFD + CI component and CFD + CPC component can achieve better feature response and semantic segmentation performance than Fig. 8(a) and (b). Finally, by combining the CFD, CPC, and CI components, the model achieves the most accurate feature response in each category, which leads to the best semantic segmentation results.

3) Effectiveness of CPC Module: To verify the effectiveness of the CPC module, we implement other causal feature alignment methods, such as global-wise adversarial training GAT and class-wise adversarial training FADA. From Table VIII, GAT improves the baseline to 75.79% on OA, 68.10% on MA, and 52.74% on mIoU, respectively, which indicates the effectiveness of the global-wise adversarial training in narrowing domain shift. FADA obtains the performance with OA, MA, and mIoU values of 76.82%, 69.90%, and 53.45%, respectively, by considering class-wise feature alignment. Compared with the above methods, CPCA achieves the best performance with OA, MA, and mIoU values of 77.00%, 71.31%, and 56.32%, respectively, which demonstrates the superiority of the class-wise contrastive training to global-wise and class-wise causal adversarial training.

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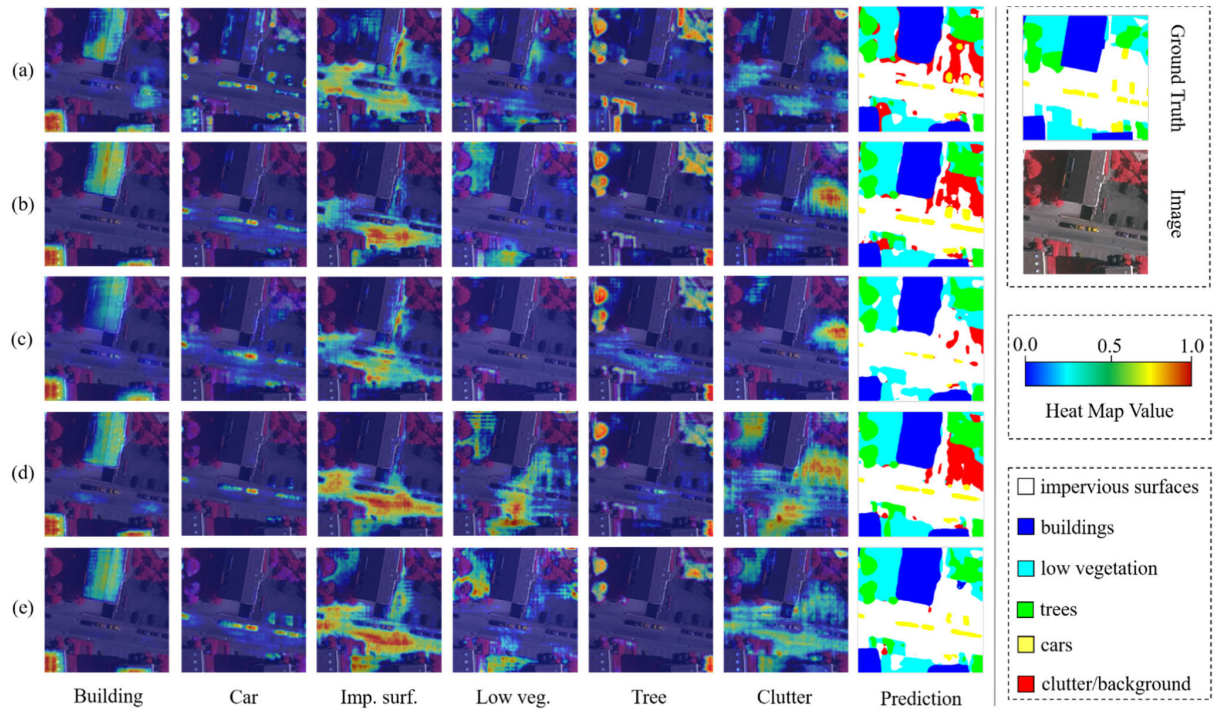


Fig. 8. CAM of each component on the POT (IR-R-G) → VAI (IR-R-G) task. (a) CAM of the causal encoder of the model with CFD components. (b) CAM of the model with CPC components. (c) CAM of the causal encoder of the model with CFD + CI components. (d) CAM of the causal encoder of the model with CFD + CPC components. (e) CAM of the causal encoder of the model with CFD + CPC + CI components.

TABLE VIII

ABLATION STUDIES OF DIFFERENT CAUSAL FEATURE ALIGNMENT METHODS FOR POT (IR-R-G)→ VAI (IR-R-G) TASK

Methods	F_1 score						OA	MA	mIoU
	Imp. surf.	Building	Low veg.	Tree	Car	Clutter			
Source-only	72.85	74.40	56.42	67.96	46.76	28.49	67.19	60.55	43.58
GAT (global-wise adversarial training)	**81.96**	88.45	53.27	76.15	51.89	51.72	75.79	68.10	52.74
FADA (class-wise adversarial training)	81.18	**89.03**	57.80	77.86	56.65	44.04	76.82	69.90	53.45
Ours (contrast training)	78.40	88.32	**62.23**	**78.66**	**59.36**	**59.33**	**77.00**	**71.31**	**56.32**

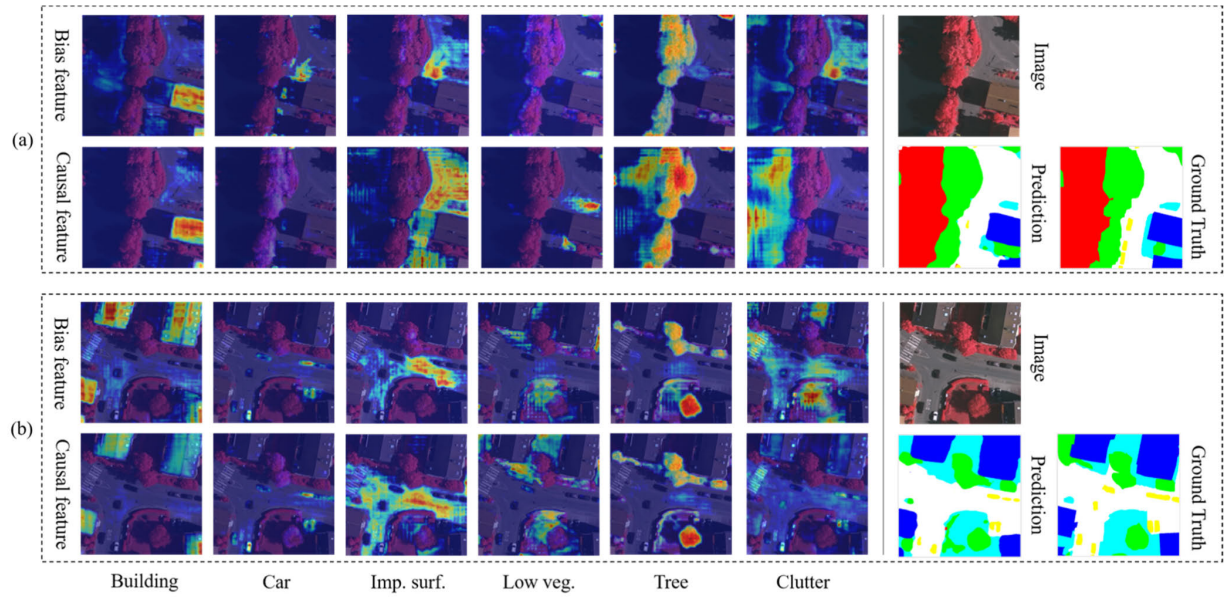


Fig. 9. Visualization of causal feature and noncausal feature on the POT (IR-R-G) → VAI (IR-R-G) task.

D. Visualization of Causal Features and Bias Features

We conduct visualization experiments on the causal and bias features of CPCA, and the results are shown in Fig. 9 with two typical remote sensing scenarios (a) and (b) in the Vaihingen dataset, where there are some differences and complementarities between the causal and bias features of

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CPCA. For example, the building causal features in Fig. 9(a) and (b) pay more attention to the global information of the building, and respond more uniformly to the features in the building area, while the bias features are more focused on the information of the local details and respond more to the information of the surrounding geo-objects. The causal and bias features of the Car are more complementary in scenario (a), with the causal features corresponding to the car and the bias features focusing more on the tree next to the car. In scenario (b), the causal features respond more accurately to the car and also consider the co-occurrence relationship between cars and trees, and impervious surfaces. For the categories of impervious surfaces, low vegetation, and trees, their causal features respond themselves more comprehensively and accurately, whereas the bias features do not accurately capture the location of each category. As for the clutter, causal features can accurately localize its region in scenario (a), while bias features focus more on other categories. Since there is no clutter category in scenario (b), both causal and biased features respond to information from other categories, but the response of causal features is slightly lower. In summary, the causal and bias features of CPCA focus on different information about geo-objects, in which the causal features pay more attention to information about geo-objects themselves, with accurate and comprehensive responses; while the bias features focus on the local and chaotic information about geo-objects with less precise and scattered responses.

V. Conclusion and Future Work

In this article, we meet the UDA semantic segmentation challenge from a causal view and propose a novel CPCA method. CPCA assumes that each source- and target-domain remote sensing image is composed of causal and noncausal factors. Since it is difficult to precisely separate causal and noncausal factors from the original input, we attempt to disentangle the domain-invariant features from the source and target domain images by disentangled representations learning, and force the domain-invariant features to replace causal factors by satisfying the following three principles: 1) causal factors should be separated from the noncausal factors; 2) the causal factors of source and target domains are domain invariant; and 3) intervening on the noncausal factors will not affect the causal factors to make decisions on labels. Through the above attempts, the CPCA can build an invariant causal mechanism between source and target domains, which enhances the generalization of the model on the target domain. Comprehensive experiments demonstrate the validity, interpretability, and generalizability of the proposed method.

Since not all causal information is most critical in domain adaptation tasks, in the future, we aim to further explore the sufficient and necessary causes rather than all possible causes.

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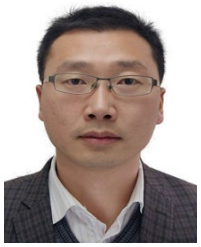
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